Finite-depth Quasi-Phase Matching (QPM) and Converter Efficiency

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An ensemble of ^{85}Rb atoms couples optical (780 nm) and microwave cavities through a blue (480/481 nm) control and UV (297 nm) dressing [2]. Optical-to-microwave conversion absorbs optical and blue photons and emits microwave and UV photons; reversing the loop reverses conversion. A separate 1188 nm lattice provides the periodic density for quasi-phase matching (QPM). Finite site width reduces the overlap of the two collective atomic profiles. We derive its effect on coherent photon-number efficiency, including atomic loss, reflection, and useful cavity ports.

I. SPATIAL OVERLAP AND LATTICE LOCALIZATION

Let $\mathbf{u}, \mathbf{v}$ be normalized optical/blue and UV/microwave excitation profiles, including relative coupling magnitudes and phases. Their amplitude overlap is

$$Q = |\mathbf{u}^\dagger \mathbf{v}|, \qquad \mathbf{u}^\dagger \mathbf{u} = \mathbf{v}^\dagger \mathbf{v} = 1, \qquad 0 \le Q \le 1. \tag{1}$$

For matched uniform envelopes, $Q = |\langle e^{i\Delta\mathbf{k}\cdot\mathbf{r}}\rangle|$, averaged over the atomic density, with $\Delta\mathbf{k} = \mathbf{k}_{780} + \mathbf{k}_{\mathrm{blue}} - \mathbf{k}_{\mathrm{UV}} - \mathbf{k}_\mu$. Wavevectors include propagation directions. Thus Q^2 is a useful-mode projection; conversion also depends on the coupled response and losses.

For lattice period $d = \lambda_{\mathrm{L}}/2$, define $k_{\mathrm{L}} = 2\pi/\lambda_{\mathrm{L}}$ and $K = 2k_{\mathrm{L}}$. Near a minimum of depth U, the harmonic approximation gives [1]

$$V(x) = U\sin^2(k_{\mathrm{L}}x) \simeq U k_{\mathrm{L}}^2 x^2, \qquad E_{\mathrm{R}} = \frac{\hbar^2 k_{\mathrm{L}}^2}{2m}, \qquad \omega_t = \frac{2\sqrt{UE_{\mathrm{R}}}}{\hbar}. \tag{2}$$

At temperature T, the position variance and density harmonics are

$$\sigma_x^2 = \frac{\hbar}{2m\omega_t}\coth\!\left(\frac{\hbar\omega_t}{2k_{\mathrm{B}}T}\right),$$
$$b_n = |\langle e^{inKx}\rangle| = e^{-(nK)^2\sigma_x^2/2} = \exp\left[-n^2\sqrt{\frac{E_{\mathrm{R}}}{U}}\coth\!\left(\frac{\sqrt{UE_{\mathrm{R}}}}{k_{\mathrm{B}}T}\right)\right]. \tag{3}$$

Here n is a spatial harmonic order. At $T = 0$, zero-point width still gives $b_n < 1$ for $n \neq 0$; deeper trapping increases the overlap. The harmonic treatment requires localization near the minimum.

Exact matching, $\Delta\mathbf{k} = \sum_i n_i \mathbf{K}_i$, aligns the phases of occupied site centres. Independent in-site displacements then give

$$Q = \prod_i b_{n_i,i}; \qquad Q = b_{2,x}b_{2,y} \quad \text{for two second-order axes.} \tag{4}$$

For example, take optical and blue beams along $+x$, UV along $+y$, and $\mathbf{K}_i = K\hat{\mathbf{i}}$. The target $(n_x, n_y) = (2, -2)$ requires

$$k_{780} + k_{\mathrm{blue}} - k_{\mu,x} = 2K, \qquad -k_{\mathrm{UV}} - k_{\mu,y} = -2K, \qquad k_{\mu,z} = 0. \tag{5}$$

This geometry is conditional on actual wavelengths and angles. Neglecting $\mathbf{k}_\mu$, nominal 780, 1188, and 480 (481) nm give residual $\delta k_x = -0.01017$ (-0.03738) rad/μm. For Gaussian site-centre rms widths L_i, residual mismatch adds the factor $\exp[-\sum_i(\delta k_i L_i)^2/2]$; negligible cancellation requires $|\delta k_i|L_i \ll 1$. In-site factors and mode envelopes must likewise be evaluated for the actual geometry. The benchmark below assumes exact matching.

II. SIX-MODE CONVERTER RESPONSE

A. Collective modes and assumptions

Let $\hat{a}, \hat{b}$ denote the optical and microwave cavities, and $\hat{e}_j^\dagger, \hat{r}_j^\dagger$ create optical and Rydberg excitations from the prepared UV-dressed state on atom j. UV dressing mixes the ground and auxiliary Rydberg states; its effect is

included in g_m and $\mathbf{v}$. Weak excitation makes these operators approximately bosonic. Assume frozen positions, uniform blue-control magnitude, identical atomic detunings, and uniform independent damping. Absorb the blue phase into $\hat{r}_j$ and the global profile phase into $\hat{b}$, so Ω is real and $\mathbf{u}^\dagger\mathbf{v} = Q \geq 0$.

With $s = \sqrt{1-Q^2}$, decompose the microwave profile as

$$\mathbf{v} = Q\mathbf{u} + s\mathbf{w}, \qquad \mathbf{w} = \frac{\mathbf{v} - Q\mathbf{u}}{s}, \qquad \mathbf{u}^\dagger\mathbf{w} = 0, \quad \mathbf{w}^\dagger\mathbf{w} = 1. \tag{6}$$

At $Q = 1$, choose any orthogonal $\mathbf{w}$; its microwave coupling vanishes. The four collective operators are

$$\begin{aligned} \hat{P}^\dagger &= \sum_j u_j \hat{e}_j^\dagger, \quad \hat{R}^\dagger = \sum_j u_j \hat{r}_j^\dagger, \\ \hat{P}_\perp^\dagger &= \sum_j w_j \hat{e}_j^\dagger, \quad \hat{R}_\perp^\dagger = \sum_j w_j \hat{r}_j^\dagger. \end{aligned} \tag{7}$$

Uniform control, detunings, and damping preserve this two-profile subspace, so only these four atomic modes and two cavities are driven. Their coupling chain is

$$a \overset{g_o}{\longleftrightarrow} P \overset{\Omega}{\longleftrightarrow} R \overset{Qg_m}{\longleftrightarrow} b \overset{sg_m}{\longleftrightarrow} R_\perp \overset{\Omega}{\longleftrightarrow} P_\perp. \tag{8}$$

The orthogonal pair is a different excitation pattern of the same atoms. It is dark to the optical cavity but couples to the microwave cavity and blue field; atomic decay makes its response dissipative.

The interaction Hamiltonian fixes the coupling convention:

$$\frac{\hat{H}_{\rm int}}{\hbar} = -\Big[g_o\hat{a}^\dagger\hat{P} + \Omega(\hat{P}^\dagger\hat{R} + \hat{P}_\perp^\dagger\hat{R}_\perp) + g_m\hat{b}^\dagger(Q\hat{R} + s\hat{R}_\perp) + \text{h.c.}\Big]. \tag{9}$$

The collective matrix elements g_o, g_m are held fixed as Q varies. The notebooks use $g_o = G/\sqrt{2}$, $g_m = G_\mu/\sqrt{2}$ [4]; these factors are conventions, not finite-depth corrections. Here Ω is the Hamiltonian matrix element, so any Rabi-frequency convention with an extra $1/2$ must be converted. All rates, detunings, and frequencies are angular frequencies.

B. Damping and driven equations

Let κ_o, κ_m be bare cavity energy decay rates, including ports and internal loss; useful port rates obey $\kappa_{j,\rm ex} \leq \kappa_j$. Atomic amplitude decay rates are $\Gamma/2, \gamma_r/2$. Loss through explicit atomic modes must not be counted again in $\gamma_{m,\rm add}$.

Using Fourier amplitudes $e^{+i\omega t}$ and zero optical-cavity detuning, define

$$\begin{aligned} D_a &= \kappa_o/2 + i\omega, & D_e &= \Gamma/2 + i(\omega - \Delta_e), \\ D_r &= \gamma_r/2 + i(\omega - \Delta_r), & D_b &= (\kappa_m + \gamma_{m,\rm add})/2 + i(\omega - \Delta_\mu). \end{aligned} \tag{10}$$

Lowercase variables denote mean amplitudes of their corresponding operators. For optical input $a_{\rm in}$, with incident photon flux $|a_{\rm in}|^2$, the response is

$$\begin{aligned} D_a a - ig_o p &= i\sqrt{\kappa_{o,\rm ex}}\, a_{\rm in}, \\ D_e p - ig_o a - i\Omega r &= 0, \\ D_r r - i\Omega p - ig_m Q b &= 0, \\ D_b b - ig_m(Qr + sr_\perp) &= 0, \\ D_r r_\perp - i\Omega p_\perp - ig_m s b &= 0, \\ D_e p_\perp - i\Omega r_\perp &= 0. \end{aligned} \tag{11}$$

With no microwave input, $b_{\rm out} = \sqrt{\kappa_{m,\rm ex}}\, b$ up to a port phase. Thus $S_{mo} = b_{\rm out}/a_{\rm in}$ and $\eta = |S_{mo}|^2$ measure coherent photon-number conversion, not energy efficiency or added noise.

C. Atomic feedback and transfer function

Set $F = D_e D_r + \Omega^2$. Eliminating the orthogonal pair gives

$$r_\perp = \frac{ig_m s D_e}{F}\, b, \qquad \Sigma_\perp = \frac{g_m^2(1-Q^2)D_e}{F}. \tag{12}$$

Its feedback changes D_b to $D_b + \Sigma_\perp$, adding full linewidth $2\,\mathrm{Re}\,\Sigma_\perp$ and dispersion $\mathrm{Im}\,\Sigma_\perp$. This loading does not contribute to the useful output-port rate.

Eliminating all atomic amplitudes gives the exact frequency-domain reduction

$$\begin{pmatrix} A & iJ \\ iJ & B \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} i\sqrt{\kappa_{o,\mathrm{ex}}}\, a_{\mathrm{in}} \\ 0 \end{pmatrix}, \qquad A = D_a + \frac{g_o^2 D_r}{F}, \quad B = D_b + \frac{g_m^2 D_e}{F}, \quad J = \frac{Q g_o g_m \Omega}{F}, \tag{13}$$

without assuming fast atomic relaxation. Consequently,

$$\boxed{S_{mo}(\omega) = \frac{\sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, J}{AB + J^2}, \qquad \eta(\omega) = |S_{mo}(\omega)|^2.} \tag{14}$$

At fixed couplings and linewidths, A, B are independent of Q, whereas $J \propto Q$: the microwave profile's useful and orthogonal loading add as $Q^2 + (1 - Q^2) = 1$. Replacing g_m by Qg_m incorrectly reduces that total loading as well as the cross coupling. At $Q = 1$ the orthogonal pair decouples; at $Q = 0$ conversion vanishes.

D. On-resonance matching

At $\omega = \Delta_e = \Delta_r = \Delta_\mu = 0$, $A, B > 0$ and J are real. Then

$$\mathcal{C} = \frac{J^2}{AB}, \qquad \boxed{\eta(0) = \frac{\kappa_{o,\mathrm{ex}}}{2A}\frac{\kappa_{m,\mathrm{ex}}}{2B}\frac{4\mathcal{C}}{(1+\mathcal{C})^2}.} \tag{15}$$

The first two factors compare useful escape with dressed damping; the last peaks at $\mathcal{C} = 1$. At fixed A, B, reducing Q lowers conversion when $\mathcal{C} < 1$, but can improve it when $\mathcal{C} > 1$ by restoring matching. Internal loss can leave reflection even at this optimum.

For comparison with the notebooks, define

$$C_o = \frac{2G^2}{\kappa_o \Gamma}, \qquad C_m = \frac{2G_\mu^2}{(\kappa_m + \gamma_{m,\mathrm{add}})\gamma_r}, \qquad C_B = \frac{4\Omega^2}{\Gamma\gamma_r}, \qquad \eta_j = \frac{\kappa_{j,\mathrm{ex}}}{\kappa_j}. \tag{16}$$

Equivalently,

$$\eta(0) = \eta_o\eta_m \frac{\kappa_m}{\kappa_m + \gamma_{m,\mathrm{add}}} \frac{4Q^2 C_o C_m C_B (1+C_B)^2}{[(1+C_B+C_o)(1+C_B+C_m) + Q^2 C_o C_m C_B]^2}, \tag{17}$$

$$\mathcal{C} = \frac{Q^2 C_o C_m C_B}{(1+C_B+C_o)(1+C_B+C_m)}. \tag{18}$$

Port fractions multiply efficiency only at fixed total linewidths; changing port coupling requires updated cooperativities. Propagation and detection losses further multiply this port-to-port result.

III. NUMERICAL BENCHMARK AND LOSS BUDGET

A. Lattice and motional scales

Take ^{85}Rb, $\lambda_{\mathrm{L}} = 1188$ nm, waist $w = 200\,\mu$m, and full retroreflection. Adopt the calibration 3 W per beam per axis $\leftrightarrow U/k_{\mathrm{B}} = 15\,\mu$K, without a polarizability calculation. For $E_{\mathrm{R}}/k_{\mathrm{B}} = 0.0799004\,\mu$K, $T = 0$, and two exactly matched second-order axes,

$$b_2 = e^{-4\sqrt{E_{\mathrm{R}}/U}} = 0.746816, \qquad Q = b_2^2 = 0.557734, \qquad Q^2 = 0.311067. \tag{19}$$

The lattice has $U/E_{\mathrm{R}} = 187.7$, $\sigma_x = 36.12$ nm, and $2K\sigma_x = 0.7641$. Thus a deep lattice still leaves substantial phase spread at the second harmonic; two axes multiply the reductions. The trap frequency is $\omega_t/(2\pi) = 45.62$ kHz, with $\hbar\omega_t/k_{\mathrm{B}} = 2.190\,\mu$K. The ground-state benchmark requires temperature small on this scale; at $T = 1\,\mu$K the same harmonic formula gives $Q = 0.48137$.

Using this equilibrium density as a static profile also requires $\omega_t\tau_{\mathrm{resp}} \ll 1$ and unresolved motional sidebands. This is a condition on evolution during conversion, separate from the phase spread $nK\sigma_x$ retained exactly in the Gaussian average. The calculation describes the coherent response, not the inelastic motional spectrum.

B. Conversion and matching

Use zero detunings, $\gamma_{m,\mathrm{add}} = \gamma_r$, and

$$\frac{(G, G_\mu, \Omega)}{2\pi} = (15.8474565,\ 2.09170145,\ 5.52497339)\,\mathrm{MHz}, \qquad \frac{(\kappa_o, \kappa_m, \Gamma, \gamma_r)}{2\pi} = (10,\ 5,\ 6,\ 0.1)\,\mathrm{MHz}. \tag{20}$$

Dividing every rate and frequency by 2π leaves efficiency unchanged. For unit port fractions:

Model	On-resonance efficiency
Matched profiles, $Q = 1$	82.0957%
Naive scaling, $Q^2\eta(Q = 1)$	25.5373%
Only $G_\mu \mapsto QG_\mu$; orthogonal pair omitted	51.5929%
Six modes, $Q = 0.557734$	**47.0824**%

Here $\Sigma_\perp/(2\pi) = 0.147394$ MHz, giving added full linewidth 0.294787 MHz. The dressed responses are $A/(2\pi) = 5.20468$ MHz and $B/(2\pi) = 2.76394$ MHz; $\mathcal{C}$ falls from 0.61947 at $Q = 1$ to 0.19270, worsening matching. With other parameters fixed, port fractions $\eta_o = \eta_m = 0.98$ give 45.2180%; $T = 1\,\mu$K instead gives 38.15% at unit ports. Setting only $\gamma_{m,\mathrm{add}} = 0$ gives 48.54% at $T = 0$.

The slowest amplitude decay rate of the six-mode drift matrix is $2\pi \times 1.4533$ MHz, so $\tau_{\mathrm{resp}} = 0.1095\,\mu$s and $\omega_t\tau_{\mathrm{resp}} = 0.0314$. This supports frozen motion for this benchmark; a narrower converter response requires a new check.

C. Reflection and dissipation

With the drive convention of Eq. (11), $a_{\mathrm{out}} = a_{\mathrm{in}} + i\sqrt{\kappa_{o,\mathrm{ex}}}\,a$. For incident flux $\Phi = |a_{\mathrm{in}}|^2$, define

$$\begin{aligned} \mathcal{R}_o &= \left|1 + i\sqrt{\kappa_{o,\mathrm{ex}}}\,a/a_{\mathrm{in}}\right|^2, \quad \mathcal{L}_{\mathrm{at}} = \frac{\Gamma(|p|^2 + |p_\perp|^2) + \gamma_r(|r|^2 + |r_\perp|^2)}{\Phi}, \\ \mathcal{L}_o &= \frac{(\kappa_o - \kappa_{o,\mathrm{ex}})|a|^2}{\Phi}, \qquad \mathcal{L}_m = \frac{(\kappa_m - \kappa_{m,\mathrm{ex}} + \gamma_{m,\mathrm{add}})|b|^2}{\Phi}. \end{aligned} \tag{21}$$

The passive model obeys $\mathcal{R}_o + \eta + \mathcal{L}_{\mathrm{at}} + \mathcal{L}_o + \mathcal{L}_m = 1$. The unit-port benchmark budget, in percent of incident flux, is

Channel	$Q = 1$	$Q = 0.557734$
Microwave output	82.0957	47.0824
Optical reflection	3.4748	37.3236
Total atomic dissipation	12.7875	14.6524
Added microwave dissipation	1.6419	0.9416

The orthogonal pair dissipates 2.7759%, included in the atomic total. Of the 35.0133-point conversion reduction, 33.8487 points appear as increased reflection. Mode mismatch mainly changes matching here; its penalty is neither $1 - Q^2$ nor the loss directly dissipated in the orthogonal pair.

IV. VALIDITY AND NOTEBOOK CONVENTIONS

Anharmonic motion, resolved sidebands, inhomogeneous controls or detunings, and additional independent optical channels require an extended response. The notebooks and direct matrix solutions check the six-mode reduction, $Q = 1$ limit, benchmark, and flux conservation; they do not validate the geometry or supply a microscopic origin for $\gamma_{m,\mathrm{add}}$.

In `Abhishek_Converter_QPM.nb`, set $\epsilon = 1$ and $\gamma_{m,\mathrm{add}} = \gamma_r$ (the latter is implicit). Its κ, κ_μ mean κ_o, κ_m; the calculator's $G_{\mathrm{opt}}, G_{\mathrm{mm}}$ mean G, G_μ. The notebook uses x, z for the two axes called x, y here. Since `ConverterEfficQ` retains its forcing amplitude `Ein`,

$$S_{mo} = \sqrt{\eta_o\eta_m}\,\frac{\texttt{ConverterEfficQ}}{\texttt{Ein}}, \qquad \eta = \eta_o\eta_m\,\texttt{NormalizedEfficiencyQ}. \tag{22}$$

For real nonzero drive, `NormalizedEfficiencyQ` $=$ `EfficiencyQ`/`Ein`^{2}. `EfficiencyQWithPorts` multiplies the unnormalized expression, so use `Ein=1` or divide its result by `Ein`^{2}. `SigmaPerp` and `OnResonanceEfficiencyQ` implement

Eqs. (12) and (17), the latter at unit ports. Use `parmsScreenshot` for Eq. (20); `parmsOriginal` retains the original parameters. Keep Q symbolic until applying `qRule`.

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[1] I. Bloch, J. Dalibard and W. Zwerger, *Rev. Mod. Phys.* **80**, 885 (2008), Sec. II. doi:10.1103/RevModPhys.80.885.

[2] A. Kumar et al., *Nature* **615**, 614–619 (2023). doi:10.1038/s41586-023-05740-2.

[3] A. Kumar et al., *Quantum-limited millimeter wave to optical transduction*, arXiv:2207.10121 (2022), Supplement B.1–B.3. Full text and supplement.

[4] Baseline: `Abhishek_Converter.nb`; QPM implementation: `Abhishek_Converter_QPM.nb`. Code and calculator: AVKGitHub1/transduction_calcs, commit `30da8841b3c9`. The Q extension is derived here.